Name what is doing the pushing or pulling

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Interaction / ES: interacción / PT: interação

Every force has a source (what pushes or pulls) and a receiver.

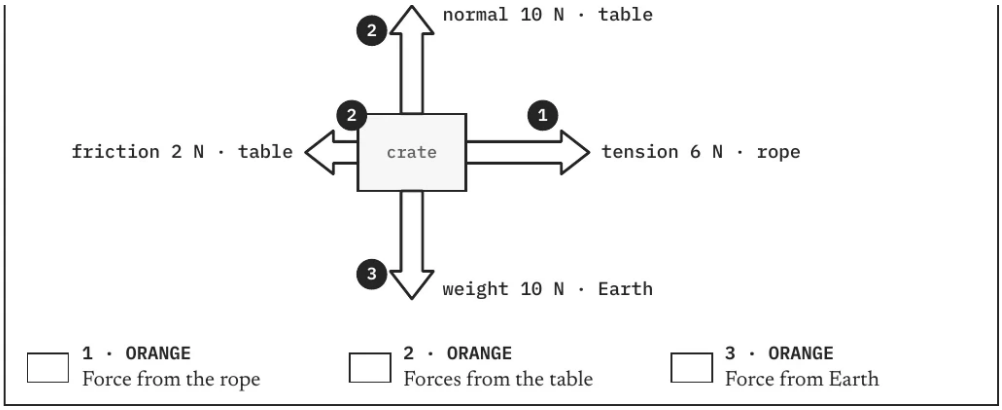
Also on your note page: Contact force, Non-contact force, Mass vs. weight.

B READ THE RELATIONSHIP

An interaction has two objects. One object exerts a force on the other.

Weight = mass × gravitational field strength

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg object rests on a level table. Find its weight and name the force that supports it.

Given m = 3 kg g ≈ 10 N/kg

Rule W = m × g

Work W = 3 × 10 = 30 N down table pushes up equally

Answer 30 N normal force 30 N up

E SAVE FOR THE FINAL MEETING

If XJ's path ever bends or speeds up, something is pulling on it. Naming the source of every force (out there, mostly gravity) is how analysts explain a change.

Read the physics: Interaction

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Ask: Which object receives the force? What other object pushes or pulls it? A rope pulls by contact. Gravity pulls even without contact.

02 HOW IT WORKS

Mass is measured in kilograms. Weight is a force in newtons. On Earth, multiply mass by gravitational field strength to find weight. Earth pulls downward; a table pushes perpendicular to its surface.

03 WHAT TO CHECK

A rope may carry part of a crate's weight. If the crate stays at rest, all upward forces together balance the downward force.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg object rests on a level table. Find its weight and name the force that supports it. 30 N normal force 30 N up.

05 CONNECT TO XJ

If XJ's path ever bends or speeds up, something is pulling on it. Naming the source of every force (out there, mostly gravity) is how analysts explain a change.

SAY IT BEFORE YOU WRITE

Who exerts the force, and on what object? The ___ exerts a force on the ___ toward ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Interaction / ES: interacción / PT: interação

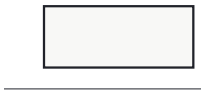
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, interaction means …	Earth pulls on a crate. Which two objects are interacting? The two objects are ___ and ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw Earth and a crate. Put weight on the crate; label source, receiver, direction, and N.

Labels you can use: source · receiver · contact.



Tell a partner: The ___ exerts a force on the ___ toward ___.

On assignment: Interaction

SECTION E / BACK / FINAL MEETING ONLY Name _____

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Simulation engineer**

Prepare the force layer for XJ. Name a source and receiver for each arrow you keep. Put motion on a separate layer.

☐ **Science graphics editor**

Sketch the graphic viewers will see. Have your producer challenge the source of each arrow before you approve its label.

Word bank: source · receiver · contact.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”